SUP'COM

المـــدرســـة العليـــا للمـــواصـــلات بتونـــس
ÉCOLE SUPÉRIEURE DES COMMUNICATIONS DE TUNIS

University of Carthage

End of Studies Internship

Submitted to the Higher School of Communications of Tunis
In fulfillment of the requirements for the award of

Engineering Degree in Telecommunications
Major :
Management and Technological Inoovation

Indoor localization system based on autarkic wake-up receivers

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Academic year: 2019/2020

ACKNOWLEDGEMENT

It is with great pleasure that I reserve these few lines in gratitude and appreciation to all those who contributed to the elaboration of this work.

I dedicate this project to my parents and my brothers. I hope that this achievement will complete a part of the dream that you had for me all those years, when you chose to give me the best education you could.

I would like to express my sincere gratitude to my supervisors at HTWK Leipzig, Prof. Dr. Ing.- Faouzi Derbel and M.Sc. Robert Fromm for providing guidance and support since my first day. Their advice were my guide to accomplish each step of the project. It has been an honor and a pleasure to work with them.

I would also like to thank my academic advisor, Prof. Rached Hamza, for his personal attention and for being always available to make such an accomplishment possible

Finally, my deepest gratitude is addressed to the whole pedagogical team of the Higher School of Communications of Tunis (Sup'Com) and especially my professors. They are the ones that I owe all my technical and communicational skills and taught me a lot about how to become a better engineer from the beginning of my studies until the last day.

ABSTRACT

The Internet of Things (IoTs) is an emerging technology intended to revolutionize the world and provide Wireless Sensor Networks with autonomous, ubiquitous and widespread computing. The acquisition of sensor node information and the regulation of "things" are typically associated with IoTs. The lack of location data for these sensor nodes, however, undermines the IoT network's intelligence. This work is therefore driven by recent developments in the two main areas of WSNs, namely indoor localization and IoTs. The main objective of this project is to provide accurate indoor localization system while maintaining energy conserved within sensors. This system is beneficial in terms of objects localization in closed areas such as supermarkets, offices and at home. The technology behind this project is Wireless Sensors Network.

Keywords : Indoor localization, WSN, Wake up receivers, Decawave.

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Glossary

WSN: Wireless Sensor Network

IoT: Internet of things

GPS : Global Positioning System

MAC : Medium Acess Control

IPS : Indoor Postioning System

NLoS : Non-Line-of-Sight

LoS : Line-of-Sight

IR : Infrared

RF : Radio Frequency

RFID : Radio Frequency Identification

WLAN : Wireless Local Area Network

APs : Access points

ISM frequency : Industrial, Scientific, and Medical frequency

ISM band : Industrial, Scientific and Medical band

WuRx : Wake-up receiver

ODIL : On demand indoor localization

RTLS : Real-time location system

DWM1001 : Decawave module

PCB : Printed circuit board

TXD : Transceiver

RXD : Receiver

GND : Ground

LPM : Low power mode

SMA : SubMiniature version A

Introduction

Localization is the method of deciding the locations of entities, devices, and other objects. Recently, it has been an active research field in which much of the research focuses on using established technology to tackle the question of determination of roles. Depending on the context in which the positioning is performed, positioning can be divided into two types: outdoor positioning and indoor positioning. Indoor positioning is carried out inside buildings, while outdoor positioning is carried out outside buildings (e.g., houses, hospitals, and malls). Different applications can require various types of technology for positioning that suit their needs and constraints. The Global Positioning System (GPS), for example, is a suitable and appropriate technology.

Wireless sensor networks (WSNs) have become an attractive technology to researchers with a considerable interest in terms of indoor localization. A growing large-scale field especially after the GPS has shown remarkable disabilities inside buildings because of the impossibility of penetrating all kinds of obstacles by it's signals (e. g. : walls).

WSNs are critically resource constrained by limited power supply, memory, processing performance and communication bandwidth. In fact, sensor's batteries are limited in terms of energy capacity which influences directly it's lifetime. Transmission subsystem has been proved to be the main responsible for battery utilisation. Therefore settling strong protocols and algorithms to ensure energy efficiency within the network is the solution to ensure a longer lifetime of the batteries so a better durability of the sensor performance. Thereupon, putting the energy consuming components in sleep mode when they are not up to work seems to be interesting to preserve energy.

Report outline

This report is divided into four chapters detailed as follows :

The first chapter introduces the global context of this project including the academic context, a presentation of the host University "Leipzig University of Applied Sciences" and the main goals to be achieved.

The second chapter summarizes the state of the art of technologies and approaches on which this work is based.

The third chapter illustrates the realization of the projects goals including hardware and the software parts of the developed Indoor Localization Platform.

The fourth chapter lists the steps followed to test and validate the whole project parts in order to be ready for manufacturing.

Chapter 1

General context

1.1| Motivation

The number of Internet-connected everyday objects grows thanks to the Internet of Things (IoT). The trend with this new paradigm that combines smart embedded devices, individuals, and systems is that we will see a change in the notion of what it means to be on the Internet" over the next few years. Wireless Sensor Network (WSN) technology is an important component of IoT in this context, since it consists of a collection of sensor nodes interconnected through wireless channels, capable of providing real-world things with digital interfaces. Moreover, as it occurs in IoT, specific WSN applications can be applied to a wide variety of domains like health, agriculture, logistics, wearable computing and others. However without distinguishing the new features that characterize this denomination, recent works considered as WSN applications are called IoT applications. In the future of IoT, wireless sensor networks have a fundamental role to play. The information collected from a monitored field can be communicated by these networks of devices through wireless links. The data collected is then transmitted across a number of nodes known as sink nodes. Finally, the data is transmitted via a gateway to a remotely located server via the internet or connected to other networks. Remote sensing devices are usually battery-driven, so their performance is sensitive to battery life being limited. Efficient management of energy deserves much of the attention in the WSNs. Routing protocols designed for WSNs must therefore effectively tackle these issues in order to enhance the lifetime of the network

1.2| Academic context

This internship is part of my journey as a 3rd year engineering student at the Higher School of Communication of Tunis, Sup'Com. It's a great opportunity to get use of the therorical background acquired througout the academic studies and apply it on a professional scale.

I realized my internship within Leipzig University of Applied Sciences (HTWK).

1.3| Presentation of the hosting university

With 6,255 enrolled students, the HTWK Leipzig is the second largest Applied Sciences University in the State of Saxony and one of the largest in Germany. The University was established on the 15th of July 1992. The roots of the University extend as far back as the 18th century, when its earlier predecessor, named the Academy of Painting, Drawing and Architecture, was founded. The most recent precursors of the HTWK Leipzig University include the Leipzig Technical University (Technische Hochschule Leipzig) which is established in 1977 after merging the Leipzig School of Civil Engineering (Hochschule für Bauwesen Leipzig) and the Leipzig School of Engineering (Ingenieurhochschule Leipzig). The good reputation of the university returns to its scientific activities by delivering practical results through focused research by forging partnerships with industry and government.

A mixture of practical teaching and application-driven study is provided by Leipzig University of Applied Sciences (HTWK Leipzig). Thanks to its extremely wide range of engineering and technical disciplines, the university stands out among the higher education institutions in the region. HTWK Leipzig is involved in a wide range of teaching and research programs in the fields of economics, social science and culture. Long-term and wide-ranging partnerships with companies not only in the region but throughout Germany

and Europe, as well as numerous externally funded research projects, demonstrate the university's high standard of research excellence. With third-party funding amounting to more than 10 million Euro per year, HTWK Leipzig is one of the top-ranking universities of applied sciences in Germany.

1.4| Project goal

During my internship, I have been implicated in the research project entitled "Indoor localization system based on energy autarkic wake up receivers", supervised by Prof. Dr. Ing. Faouzi Derbel and Robert Fromm. The main goal of this project is preserving energy within WSN through setting an energy-efficient on-demand protocol. In this protocol, the wireless nodes are maintained in a low power idle listening mode using low power Wake-Up receiver and they are activated only when a localization request occurs. This method permits to optimize the activity time of the wireless node and extends its battery lifetime.

The steps to be taken throughout the internship are:

- Summary of state-of-the-art study : different energy conservation approaches and techniques used in WSN in general and discusses their efficiency within the requirements of the indoor localization systems.
- Proposing methodology and describing the hardware and the software parts of the developed Indoor localization system based on wake-up revceivers.
- Implementing a proof of concept of the methodology proposed.

Chapter 2

State of the art

2.1| Introduction

The main goals fixed, the project was based on the state of the art and the identification of the newest researches and technologies in this domain. In this chapter we will discuss the anterior works, the experiences, the results that they led to on which we are based in order to choose the techniques and technologies that will later be used as a baseline.

2.2| Wireless Sensor Networks

A WSN is a large number of autonomous and self sufficient nodes with different tasks. A node is a small computer with sensing and communicating capabilities. The interconnection between those nodes is maintained through wireless communication using transceivers.

The main objective of the sensors within the WSN is to gather information through sensing interfaces and send them to central computing node where they will be processed later. Sensors have become sophisticated and tremendously contributing to this field of research especially in the development of low-cost solutions. Meanwhile, low power

consumption is one of the challenging issues. The nodes are becoming battery-powered in order to overcome high costs and maintain a dynamic and flexible installation. A problem that arises here which consists in the limitation of longevity of the batteries.
Radio transmissions are greedy in terms of energy conception, that's why latest researches are aiming to ensure energy efficiency within WSNs.

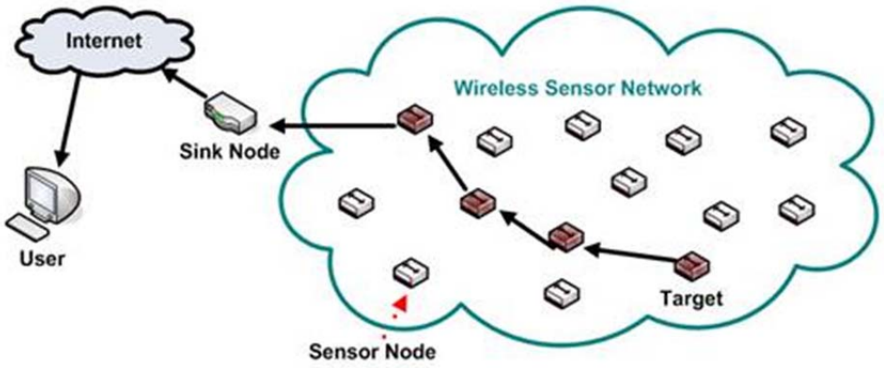


Figure 2.1 – Wireless Sensor Network model

2.3| Indoor positioning systems

2.3.1 What are IPSs?

An indoor positioning system (IPS) is a system that determines the location of a person or an object in an indoor environment continuously and in real-time and has different applications. IPSs can be used for various private home applications, including object identification and tracking, support for elderly and disabled people in their everyday operations, and medical monitoring for vital signs and emergencies.

2.3.2 Why an IPS?

There are several features that differentiate indoor positioning from outdoor localization. Indoor environments seem more complicated compared to outdoor environments,

since there are many items (such as pieces of equipment, walls, and people) that represent signals and contribute to problems with multi paths and delays. Indoor environments often usually rely on non-line-of-sight (NLoS) propagation due to the presence of different artifacts through which signals do not pass directly from an emitter to a receiver, creating inconsistent time delays at the receiver, see figure below. In addition, the presence of artifacts contributes to high attenuation and signal dispersion.

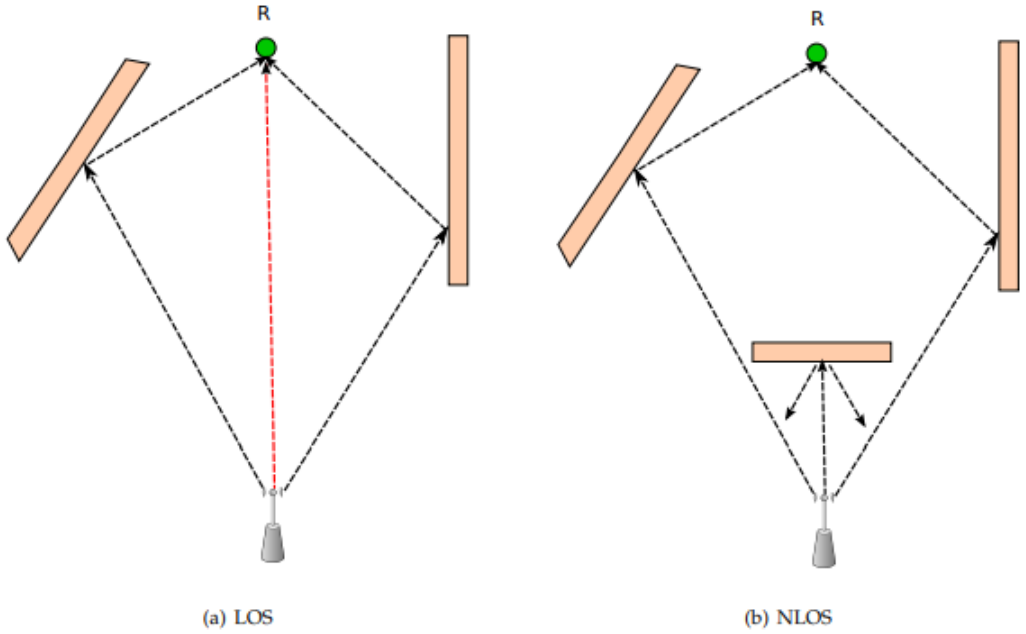


Figure 2.2 – (a) Line-of-sight (LoS) vs. (b) non-line-of-sight (NLoS) [1]

There are some features of indoor environments, on the other hand, that facilitate positioning [1]. In order to promote positioning within a limited coverage area, for instance, many factors will help: predetermined facilities, corridors, entrances and exits, small gradients of temperature and humidity, and slow air circulation. Also, indoor environments are less dynamic because objects move at a slower speed within them.

2.4| Technolgies of localization

In this segment, the widely used indoor localization technologies are briefly discussed to provide a general insight into their operational capabilities and limitations for the purpose of system location in an indoor environment application.

2.4.1 Infrared (IR)

IR light is an electromagnetic radiation having wavelengths that are more than the noticeable lights. The wavelength starts from 300 GHz to 430 THz frequency. An IR light is either produced or absorbed by molecules as vibration movements change. The key drawbacks of IR are that it works when there is a line of sight transmission between the sender and the signal receiver, the transmission distance is typically very limited and presents real issues of privacy and security as mentioned in [2] and [3]. The benefits of IR positioning systems are that they quite reliably deduce positions. IR senders are compact, not heavy; installation does not require much time and maintenance is relatively cheap.

2.4.2 Radio Frequency (RF)

RF is the rate of oscillation between the frequency of radio waves from 3 kHz to 300 GHz and the alternating current that transmits the radio signal. RF is the use of radio between the transmitter and the receiver (wireless communication). The ability of radio waves to pass through solid media, including human bodies, offers broader coverage in terms of signal propagation for RF-based technology. Indoor positioning systems for RF technology need less hardware and can re-use existing RF technology, e.g. WLAN Access Points. RF has a precision of 1 to 3 meters and has no direct line of sight. They are affected by multipath, fading, and people moving in the environment [17].

2.4.3 Radio Frequency Identification (RFID)

This is when RF electromagnetic fields are used in a non-wired medium to transmit data to identify and track IC tags (i. e. tags which have the function of wireless communication by radio wave) between transmitter and receiver. The technology is light and the IC tags are quite small, making them portable. The basic issue is the need for the

implementation and maintenance of many infrastructures where the RFID localization method is used.

2.4.4 Wireless Local Area Network (WLAN)

WLAN uses either RF or IR light in transmitting signals from sender to receiver. The two major families of WLAN technologies are not compatible due to differences in the use of RF spectrum. The 2.4 GHz Industrial, Scientific, and Medical (ISM) frequency band and the 5 GHz Unlicensed National Information Infrastructure (UNII) frequency band are those using the RF spectrums. The technology is easily influenced by human body movement and direction, overlapping access points (APs), wall thickness and material, door opening and closing, mobile units, etc. WLAN technology has the ability to improve the position estimation of stationary and mobile devices or individuals, but for localization purposes, it is reasonably precise. They are cost effective and use the resources already available for localization.

2.4.5 Bluetooth (IEE 802.15.1)

This is a non-wired technology for sending and receiving data (audio, video, other files, etc.) between devices such as mobile sets, computers, etc within a small range by utilizing short wavelength radio communication in the ISM band within 2400 to 24800 MHz. Bluetooth has the potential of linking many deices simultaneously and solves the difficulties of synchronization. Bluetooth technology is reusable that is it uses infrastructures that already have built in Bluetooth chips. It is cost effective and consumes very little energy while in operation. The basic disadvantages are that of its operational range of 2 m to 3 m with a waiting time of 20 s.The common setbacks of RF affect it.

2.4.6 Ultra Wideband (UWB)

This is a radio technology which in operation consumes very little energy. It operates over a limited distance with a bandwidth greater than 500 MHz, and its transmission uses a very high proportion of radio spectrum[4]. UWB includes a limited sound range, level and similar. It has the ability to sieve reflected signals from the main signals to provide better precision, no line of sight needed, no multi-path effect, low interference,

better diffusion power, not costly and sizeable.

2.4.7 Vision Technology

In an indoor environment, Vision technology uses mounted cameras on ceilings and walls to capture photos that are sent to a central server for the purpose of identifying and monitoring people or objects inside a building. In an incredibly large area, a cheap camera can be used. The disadvantages of this technology are anonymity, unreliability in changing environmental circumstances such as light intensity, weather conditions, and also when the artifacts to be located and monitored are in motion and several.

2.4.8 Wi-Fi (IEEE 802.11b)

The indoor localization can be also developed using the existing infrastructure of the Wireless Local Area Network (WLAN). By exploiting the multiple Wi-Fi Access Points (AP), which are already installed in different places inside the building, the location of the mobile devices is estimated using RSSI-based techniques. The WiFi-based indoor localization systems are very encouraging since they does not require neither additional infrastructure equipment nor add-ons in the mobile devices since most of them are already equipped by the Wi-Fi IEEE 802.11 (2.4/5) GHz modules. Typically, the computations are carried out in the mobile device, thus there is no need to setup specific software in the APs.

Power consumption poses a big problem for indoor localization systems focused on WiFi. The mobile devices' battery capacity is limited and the high level of WiFi communication operation rapidly reduces the power level. Building structures such as walls often limit the Wi-Fi APs' coverage range, which affects the accuracy of localization.

2.5| Architecture of a sensor node

The sensor node is a small device composed by four basic subsystems which are: the sensing subsystem which is responsible for data acquisition, the processing subsystem

which is used for data processing, the wireless communication subsystem which is in charge for data transmission, and finally the power supply subsystem which contains battery with limited energy capacity.

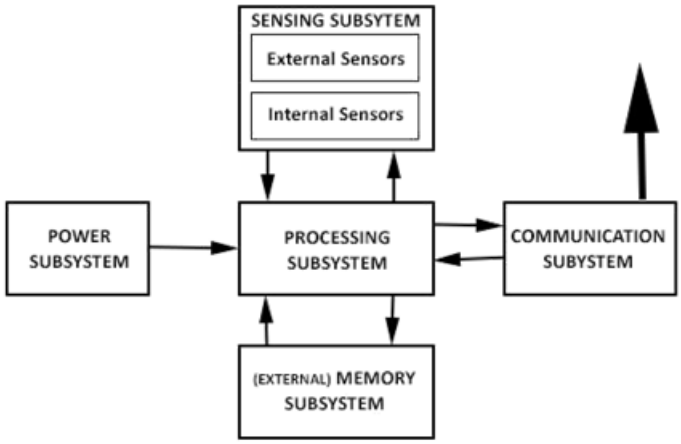


Figure 2.3 – Sensor's susbsytems

2.6| Energy conservation techniques

The main cause of energy consumption within WSN are the radio communications, therefore researches are proposing energy conservation techniques in order to extend battery lifetime of the sensor nodes. These techniques are composed into two classes which are Data-Cycling and Data-Driven.

2.6.1 Duty cycling

The Duty Cycling consists of programming the time lapses in which the sensor goes active. In other words it is an approach based on reducing the radio-activity duration of the node through shutting it down to sleep mode as long as possible. A very important point that should be mentioned is that energy consumption is high whenever the node is

transmitting/receiving data or even in idle mode. Two techniques are used to ensure this approach which are Topology control and Power management techniques.

The main idea of topology control is using the minimum number of active nodes and keeping the remaining nodes in sleep mode while maintaining the wireless connectivity. In this way the longevity of the network can be improved and the battery's power level is conserved. The criteria behind choosing which nodes to go active and which nodes to shut down can be location-driven or connectivity-driven. The location-driven protocols select the nodes that should be activated based on their positions which is known in advance. The connectivity-driven protocols dynamically activate/deactivate the network's nodes in order to ensure the network connectivity and/or complete sensing coverage.

Power management techniques are based on developing MAC protocols which are sleep/wake up and low-power MAC techniques. The sleep/wake up protocols are mainly divided into three classes which are on-demand, Synchronous, and Asynchronous. In on demand protocols, the wireless node is activated only when it is requested for communication from a neighboring node, otherwise, it remains at sleep mode. This method is highly efficient in terms of energy conservation and it is commonly used in low duty cycle or event-driven scenarios.

In synchronous mode, the activity schedule of the node is arranged according to a specific protocol such as rendez-vous method. The node is activated at certain time slots to check for potential communication requests from neighboring nodes. This method requires synchronisation between the communicating nodes so that their activity periods correspond to each other.

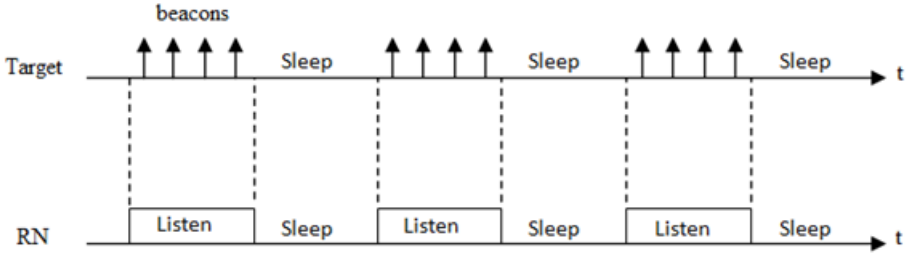


Figure 2.4 – Synchronous Mode [5]

Whereas in asynchronous mode, the activity period of the sensor node is done independently to other nodes. The activity time of the nodes should be overlapping to ensure the communication.

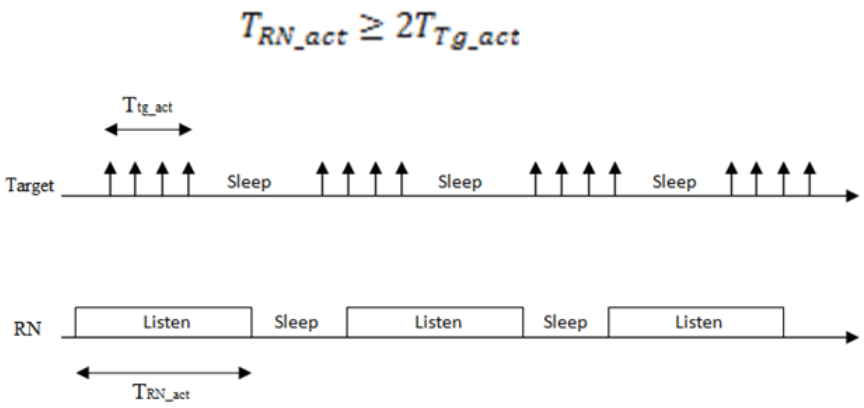


Figure 2.5 – Asynchronous Mode [5]

2.6.2 Data driven

Data driven technique is based on reducing the amount of data sent to the sink node in order to minimize energy consumption wihtin the network. Multiple methods are proposed using this technique such as conceiving the manner how data is gathered and routed within a multi-hop network and how it's processed at intermediate nodes. This method avoids the most constrained intermediate nodes in terms of energy. Data compression, aggregation and prediction are also proposed as solutions to conserve energy. In certain particular applications, the technique of data aggregation is applied to minimize the amount of data transmitted through the sensor nodes. The average value of the received data is computed by the intermediate nodes from the source to the sink. The volume of data is thus greatly decreased across the network.

In the other side, the data compression technique makes the energy consumption more efficient by reducing the amount of data sent by the source node. This latter encodes the information that it has received/sensed using different compression techniques and sends them to the sink node to be decoded and recovered.

Data prediction is another strategy that can be used to reduce energy consumption. This model predicts the values with some margin of error that can be given by the sensors. The sensor nodes and the sink interact directly by retrieving the actual sample in order to update the model if the model is not accurate enough. This approach decreases the amount of data sent by source nodes as well as the amount of energy needed for communication.

2.7| Wake-up receiver

A wake-up receiver (WuRx) is an ultra-low-power radio receiver that performs idle listening while keeping the main radio completely off. A WuRx consumes much less power than the main transceiver and only triggers an interrupt when a packet is received with a user-defined address. Embedding such a device enables better event-triggered applications where real-time behavior is required and a longer lifetime is mandatory. A WuRx is intended to monitor the channel continuously, waiting to receive a triggering event known as wake-up packet (WuPt). The latter includes the transmitter's details and in particular, the destination address. It is only sent when the transmitter has to communicate with the receiving node. The WuRx verifies whether the destination address correlates when a WuPt is received and then issues an interrupt command to the MCU. The receiver radio is activated to establish the link with the transmitter. For a duty-cycled radio to have a similar energy consumption of a WuRx, the duty cycle should be very low (i.e., 0.01 %). As a result, the latency is significantly increased.

2.8| Taxonomy of WSN-based indoor localization techniques

2.8.1 Range-based measurements models

In order to estimate the distance and/or angles between wireless nodes, range-based localization techniques rely on radio signals. Range-based measurements models are many

and the most common used models are :

2.8.1.1 Time of arrival

Time of Arrival localization solutions estimate distances between devices based on the propagation time of a Radio Frequency (RF) wave between sender and receiver. Since the speed of propagation waves is known, a corresponding distance can be determined based on the measured time. When one way TOA is used, a very accurate clock synchronization is required between sender and receiver.

2.8.1.2 Angle of arrival

In AOA, the anchor measures the arrival angle of the emitted source signal. The position of the source can be estimated by the intersection of at least two lines of bearing from the source and receiver. In other words, the anchor receives the RF-signal sent by the target node (i.e. called beacon) and measures its arrival angle thanks to built-in multiple array antenna.

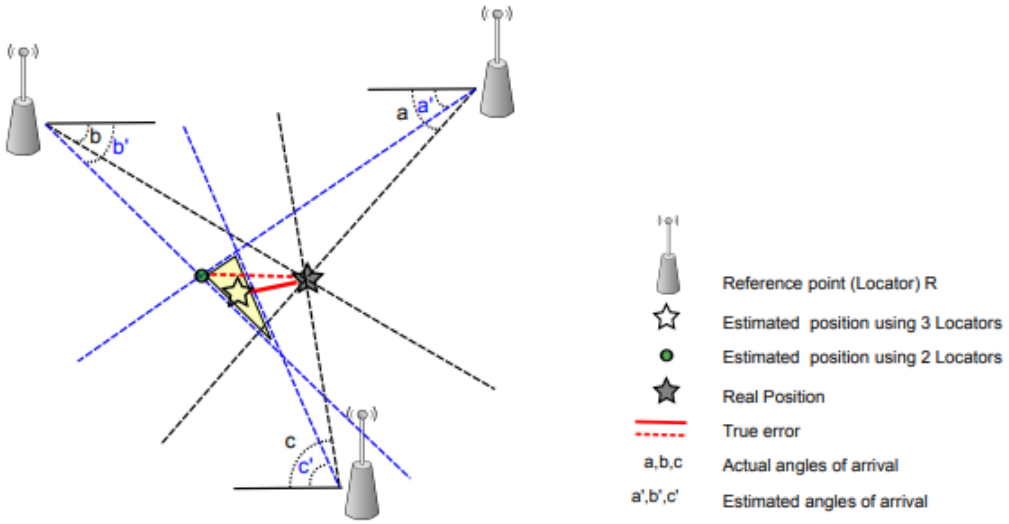


Figure 2.6 – Angle of arrival (AOA)-based algorithms [1]

2.8.1.3 Time difference of arrival

The difference of arrival of the same signal to a couple of receivers is the aim of the Time difference of arrival (TDOA). This difference is then multiplied by the velocity of propagation of signal.

2.8.1.4 Cell of origin

In the cell of origin method, the target node is estimated within a cell. A lot of anchors and cells are required within a small area in order to maintain a high accuracy.

2.8.1.5 Lateration

Lateration is a localization algorithm which is responsible of measuring distance between a target and anchors. It requires at least 3 anchors that should be non-collinear in case of 2D localization.

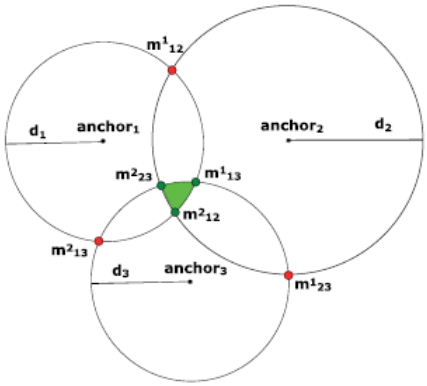


Figure 2.7 – Lateration algorithm [6]

The target is the center of gravity of the intersection points.

2.8.1.6 RSS-Based Algorithms

In RSS-based algorithms, the tracked target measures the signal strength for received signals from multiple transmitters in order to use signal strength as an estimator of the distance between the transmitters and receivers. This way, the receiver will be able to estimate its position relative to the transmitter nodes. Although RSS is sensitive to

multipath interference and a small-scale channel effect that causes a random deviation from mean received signal strength, it is used frequently with unrealistic assumptions.

2.8.2 Range-free approach

Without the need to quantify metrics such as distances or angles between a set of reference nodes and the target, localization in the range-free method is carried out. In the range-free approach, there are two main approaches, which are connectivity information and fingerprinting.

2.8.2.1 Connectivity information

In this technique based on their connectivity information to the target node, a set of reference nodes is chosen. Connectivity information ensures that the chosen reference nodes can for example, use their radio connectivity to detect the presence of the target node in its proximity. Algorithms such as Cell of Origin can be implemented to approximate the target's location after selecting the reference nodes. The connectivity information technique is based on some assumptions.

Since the range-free approach does not rely on precise distance measurements, all the deployed reference nodes are assumed to have the same range to the target node once is detected. Also, the reference node is considered geographically far from the transmitter (e.g. the target node) if the received radio signal strength is low, and vice versa. However, these assumptions do not always conform to reality especially in indoor environment. For this reason the localization accuracy of this technique is considered coarse-grained.

2.8.2.2 Fingerprinting

The pattern-matching, also known as fingerprinting, is a range-free localization method and has shown better accuracy than one of the connectivity data information. In the off-line process, the received radio signals are stored in a database called a radio map at selected locations within the building. After the radio map is formed, the location of the target node is estimated in the online process.
As we can see by taking signal measurements at many locations inside the building, this

approach involves a handmade setting of the radio chart (i.e. fingerprints). This practice is neither time-saving nor cost-effective. Moreover, precision depends heavily on the number of anchors deployed and the spatial density of the fingerprints.

2.8.3 Indoor localization systems' steps

Indoor localization in based on three main parts which are platform calibration, the monitoring and the position estimation. Those steps are executed as follows :

2.8.3.1 Calibration

This step has a main goal that consists in determining the signal's behavior within the building such as the attenuation factor and the noise effect.

2.8.3.2 Monitoring

This is the most important phase in the whole process and is responsible for the system's accuracy. In fact in the monitoring step, we detect the localization requests, take measurements from radio signals sent by target node and transmit obtained data later for processing.

2.8.3.3 Position estimation

In this step, the computing node collects all received data, applies a localization algorithm and provides the estimation of the target's position.

2.9| Related works

Energy consumption has been a real problem in WSN-based indoor localization systems. Many anterior works have dealt with this subject and proposed Indoor Localization Platforms with battery powered nodes and techniques to ensure energy efficiency and a longer battery lifetime.

Bluetooth Special Interest Group (SIG) has proposed a Bluetooth-based technology with a power-efficiency functionality. The BLE technology permits to optimize energy consumption in wireless sensor nodes and it has a wide field of use especially in the IoT applications. The low power consumption of BLE is achieved by the lower duty cycle, faster connection, and lower peak power. Like any Bluetooth device, the BLE node can operate either as a master or as a slave. In indoor localization scenario, the slave nodes are represented by anchors and the master nodes are represented by target nodes. To realize a connection, the slave node transmits periodically advertisement packets on the advertising channels while the master scans within a specific duty cycle. The energy consumption within the slave node depends on the transmission power and the rate of the signal emitted by this latter.

Another solution proposed to minimize energy consumption within the indoor localization platform is the reduction of the number of nodes involved during the localization process. Authors in [8] have presented Energy- Efficient Opportunistic Localization (EEOL) technique that aims to reduce participating nodes in the localization process through a probabilistic wake-up based on the opportunistic model. The target node gets to know how much nodes are nearby (in its range) by exchanging messages, computes the opportunistic wake-up probability and forward it to the nodes. The nodes decide even to wake up or remain in sleep mode based on this probability. This technique requires network density in terms of nodes which increases the system cost. Besides the simulation results have shown that the node has a maximum battery lifetime of 10 to 15 weeks when powered by a 720 mWh battery.

In [9], a solution based on scheduling algorithm was proposed that consists in reducing the number of anchors participating in the localization process. This study has shown that the difference in terms of accuracy between a low and a dense network is about 10%. An anchor becomes active when it detects that neighboring nodes are not well connected. This decision is based on a design parameter that describes the density required for good localization accuracy. This proposed algorithm has increased the network lifetime but the accuracy decreases after a period of time. Every 100,000 s, a set of nodes run out of battery and the system's accuracy decreases. The problem of this solution is that it's not cost effective.

An energy efficient algorithm based on varying the transmit power of the anchor node has been proposed by authors in [10]. The anchor sends beacon with the lowest power level transmission and keeps increasing the power until it reaches the maximum level.The idea behind the algorithm is to avoid a fixed transmit power level of anchors when broadcasting beacons and to save energy up to 67%. Also, it provides the target node more information each time they receive beacons to achieve better accuracy. The authors have not studied the whole energy consumption of the anchor node and how it is duty-cycled but it is kept processing periodically.

In [11], an experimental evaluation is presented of three commercially available UWB positioning systems: Ubisense, DecaWave, and BeSpoon. The evaluation has been done in an NLOS environment similar to an industrial warehouse with diverse equipment (robots, vehicles, railways, etc.), which perturbates UWB radio propagation. Under these conditions, the performance the three systems have shown good results in terms of position estimation but the DecaWave system has shown that it's slightly better than the BeSpoon system and significantly more reliable, in terms of accuracy and outlier content, than the Ubisense system. Decawave makes it possible ranging measurements with an accuracy of ±10 cm using two-way ranging (TWR) TOF measurements. The manufacturer estimates a real-time location accuracy for a moving tag of about ±30 cm in X and Y , using either TWR (TOF) measurements or one-way TDOA approaches. A maximum measurable range of 300 m is possible in ideal conditions
A comparison between the mentioned works is presented in the table below.

Works	Radio frequency	Energy optimization tech.	Localization tech.	Accuracy	Max. battery lifetime
iBeacon Apple [2016]	BLE	Low duty cycle	Proximity	2 m	2 Years
Xia et al.	N/A	Anchor selection duty cycle	Trilateration	1.6 m	3.5 Months
Gribben et al. [2010]	2.4 GHz	Anchor selection	RSSI-based Trilateration ML Estimator	10 m - 20m	1 Week
Farooq-I-Azam et al. [2016]	2.4 GHz	Tx power optimization	RSSI-based Multilateration	N/A	1 Day

Figure 2.8 – Comparison between the mentionned works [11]

2.10| Limitation of the existing solutions

In this section we are discussing the limitations of the previous works to emphasize the inconveniences that prevent us to use their results as baselines.

In the literature, the impact of noise on RSSI has been considered. To mitigate its effect on ranging accuracy, time-averaging technique is performed when collecting RSSI measurements. None of the presented RSSI-based ranging models has considered the indoor variation of temperature and humidity on the ranging accuracy. As presented in the literature, these two physical quantities have an impact on the RSSI variation. In fact, RSSI is sensitive to these factors which leads to fluctuating results in a way that makes the localization process inaccurate.

As shown in the previous table, the low duty cycle is the most efficient technique in terms of energy consumption with comparison to the other techniques in the literature, and it provides a battery lifetime in years scale. This year-scale longevity is due to keeping the wireless node in active mode for short periods during the cycles and in sleep mode in the rest of the time. The problem here is that the active period of the node is set in advance. If we extend the sleep period of this latter we increase its longevity but in revenge its responsiveness decreases. So duty cycling is a double edged weapon. Besides, in indoor localization scenario, at least 3 anchors must be active at the moment of target's presence in their ranges to respond to its request and carry out its position estimation. The condition that should be verified in duty cycling is that the beacons sent by the target node are received by the 3 anchors.

As mentioned in [5], the main drawback of duty cycling is that it keeps the anchors waking up periodically even there is no demand for the localization service in their ranges and uselessly draining their battery power.

2.11| Conclusion

Based on the previous works and what they led to, we have developed a new on-demand indoor localization protocol that permits to overcome the problems presented by the techniques mentioned above by keeping the wireless nodes on their lowest power mode until they receive a localization demand. Our proposed solution permits to maximize the battery lifetime, by disposing of the energy-consuming periodic idle listening of the duty-cycle technique.

Chapter 3

System design

3.1| Introduction

Since the Indoor Localization system's sensor nodes are battery-powered, it is important to apply methods of energy conservation to extend the life of the battery. Much of its battery power is consumed by the sensor node when transmitting or receiving data, and even in idle status. However, when it is in sleep mode, power consumption drops outstandingly. By switching periodically to sleep mode, the duty-cycling method helps the sensor node to decrease energy consumption. As seen above, this approach allows for a battery life of up to 2 years to be extended. Despite the usefulness of duty cycling, we have found that when it is used in indoor localization situations, there are many drawbacks. The sensor wakes up regularly and remains active for a certain time in order to detect the localization request. In general, this poses a waste of energy because there are neither activities nor localization requests. For this purpose, we have established an indoor localization technique that is on demand where the sensor nodes are held at their lowest power mode and only become active when they are needed. Our solution allows the battery life to be extended without affecting the responsiveness of the sensor nodes. In this chapter, we are presenting the specification of the proposed method, the new architecture of the indoor localization platform and the used technologies.

3.2| WuRx-based on-demand indoor localization

When the duty cycle approach is implemented in the indoor localization case, major energy loss has been observed in idle listening. As mentioned above, even though no events take place, the anchors keep waking up. We have found this issue and developed an indoor localization platform based on on-demand using the Wake-up Receiver (WuRx) technology.

The WuRx is a low power radio receiver that is added to the sensor node. Since the transceiver of the sensor node consumes energy whenever it is active, the WuRx enables to leave the anchor's transceiver off and to activate it only when needed. The used WuRx is identity-based which enables also to reduce overhearing since only needed nodes will wake up and listen to the channel. In addition, the WuRx helps to increase the responsiveness of the anchors within the Indoor localization system. When a wake-up packet (WuPt) is received, the WuRx is kept permanently active to monitor the channel and to enable the main transceiver. The WuRx must thus consume ultra-low energy. A solution for reducing the energy consumption and enhancing the sensitivity of the WuRx was proposed in [12]. The mean power consumption of the WuRx is less than 3 µA. The nodes in WuRx-based ODIL are maintained on a reactive basis, unlike the duty cycle technique, and are only involved when a localization request is received. In this scenario, the anchors will no longer wake up regularly to listen to potential beacons, but as long as they have not received any WuPt (i.e. localization request) from the targets in range, they are held at ultra-low power status.

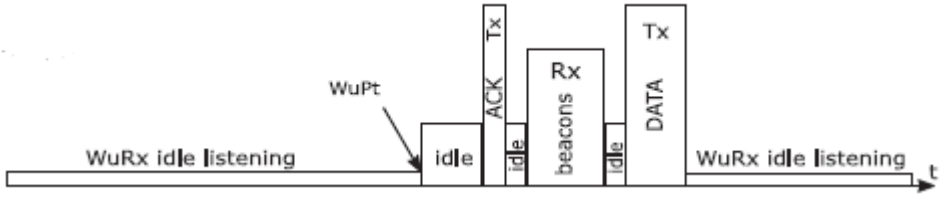


Figure 3.1 – On-demand activity of Anchor nodes in WuRx-based Indoor Localization Platform

3.2.1 The new architecture of the WSN using WuRx

The localization system is updated since a new device is introduced. The architecture of the WSN has obviously changed to satisfy the implementation of the wake-up receiver within the network. The figure below represents the new aspect of the proposed indoor localization system.

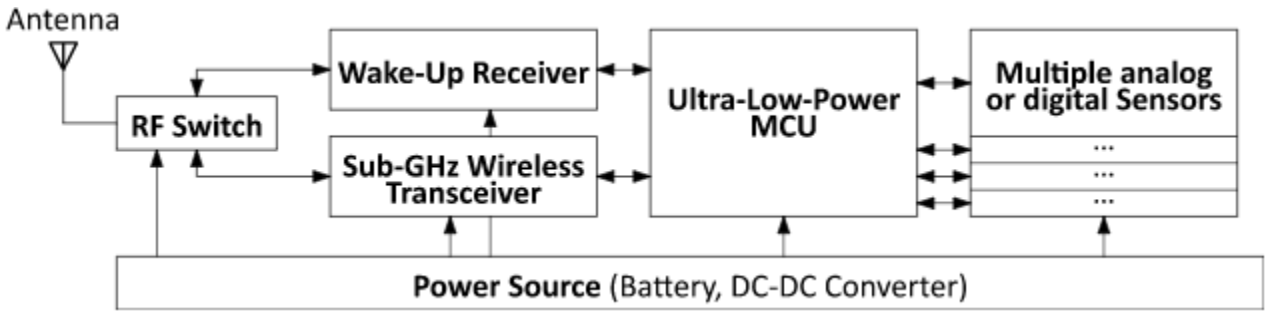


Figure 3.2 – The new concept of the WSN based on WuRx

The next figure presents the layout of the sensor node that is used in the ODIL framework based on WuRx. It is formed by two radios: the ultra-low power WuRx and the main radio transceiver with high data transmission and reception capabilities. Both radios share the same antenna output and it is up to the Microcontroller Unit (MCU) to use a radio switch to control access to them.

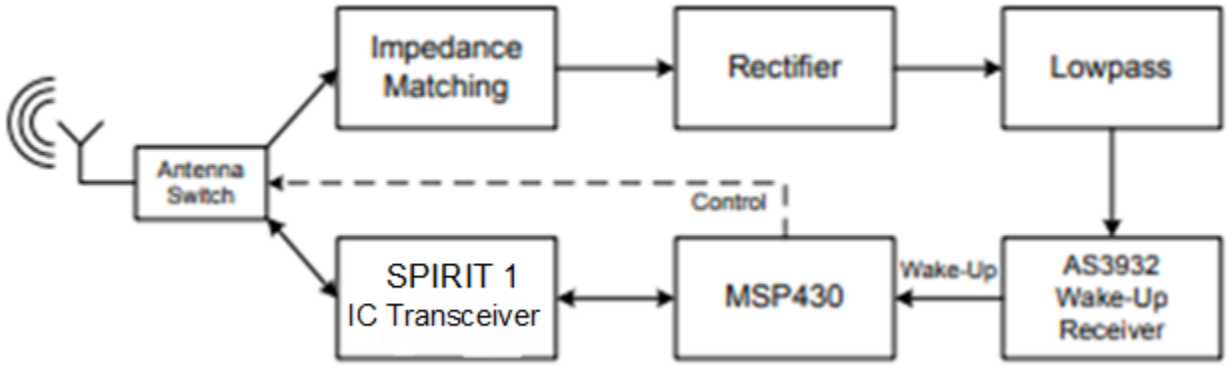


Figure 3.3 – Architecture of sensor node in ODIL system [6]

The main radio is off and the WuRx is working and has access to the antenna initially.

This mode is called low power WuRx idle listening. At this mode, the WuRx consumes only 2.7 µA by providing a sensitivity of −60 dBm. The distance of coverage (range) line of sight of the WuRx goes up to 80 m
When a WuPt is received by the WuRx, it triggers an MCU wake-up interrupt. Then the latter switches on the WuRx, activates the components of the sensor, and sets the main radio on the antenna.

3.2.1.1 Microcontroller

The microcontroller used is Texas Instruments' MSP430G2553. In its Low Power Mode 4 (LPM4), where the CPU as well as all components are disabled, it was chosen because of the current consumption of 0.5 µA. The switching time from sleep to active mode is determined by 32 µs [7]. The controller can still respond to an external interrupt on one of its I/O-pins while in sleep mode. The lower limit in RAM was defined to at least 2 KByte in order to have flexibility in testing different MAC protocols.



Figure 3.4 – MSP430G2553 Launchpad

3.2.1.2 Radio Transceiver

SPIRIT1, the IC transceiver is used as the main radio node. It is one of the low-power transceivers available in Europe in the 868 MHz ISM (Industrial, Science, Medical) band and is checked to operate correctly with the MSP430 equipment. According to the

SPIRIT1 datasheet the current consumption in active mode with +11 dBm output power is at 21 mA in transmit mode and at 9 mA in receive mode.

3.2.1.3 Wake up receiver

Low power wake-up receiver are used for keyless-go entry systems in automotives. They are build to work a long time without a battery change and therefore operate at low frequencies. The short wake-up range of about 3 meters due to inductive coupling is of no limiting factor for the keyless-go application. In our node we used the AS3932 wake-up receiver from Austriamicrosystems [14]. It consumes in one channel listening mode 2.7 µA current and has a wake-up sensitivity of 80 μV_{RMS} with a 3 dB booster as well as a high input impedance of 2 MΩ. One of the main reasons for choosing the AS3932 is, that it has an integrated correlator which compares the received signal to a byte pattern saved in a configuration register. When the two signal match, the "wake" output of the chip changes its logic state and a connected microcontroller can receive the signal on an interrupt capable input port.

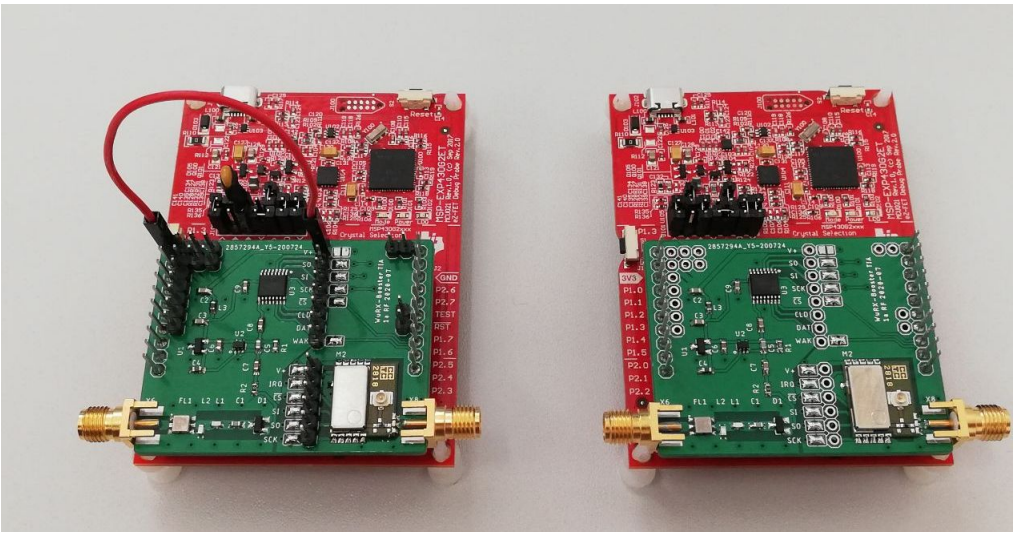


Figure 3.5 – Assembled WuRx printed circuit board and attached to a host node

3.2.1.4 Antenna Switch

In our node, there are two routes where data signals will spread from the antenna to the hardware of the node. One way is directly to the main radio transceiver, the other way,

via the analog circuit. To avoid having two antennas on one node, a low-power antenna switch was integrated to control the propagation of incoming signals. The ADG918 switch from Analog Devices has a high off isolation of 43 dB at 1 GHz and consumes less than 1 µA current.

3.3 Decawave

Based on the benchmark elaborated in the previous section between Decawave, BeSpoon and Ubisense we decided to continue with the Decawave. In this section we will dig into this technology and present its different aspects.

3.3.1 Overview

The DecaWave Module DWM1001 is a new ultra-wideband transceiver that advertises high-precision indoor pairs with errors as low as 10 cm between modules. Anchor positioning, with limited time constraints and static instruments, is a restricted version of indoor localization. Ranging is defined as one of the core functionalities of the DecaWave transceiver (measuring the distance between two nodes). DecaWave uses a time-of-flight (ToF) method to cancel out any offsets in absolute time that may be present between the nodes performing range due to the inherent difficulty in maintaining precise long-term time synchronization between nodes.

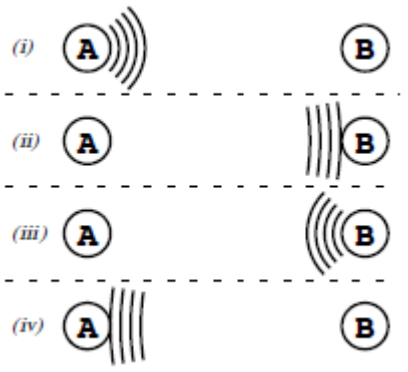


Figure 3.6 – Visualization of time-of-flight (ToF) ranging [19]

The previous figure illustrates the ToF ranging concept to determine the range between two nodes A and B. The ToF method used by the DWM1001 calculates the range based on both the two-way propagation time and a known turnaround time at node B

$$(((\ T_{iv}\ -\ T_{i})\ -\ (\ T_{iii}\ -\ T_{ii}))\ \cdot\ c/2\ )$$

instead of calculating the distance based on the immeasurable RF propagation time from A to B

$$((T_{ii}\ -\ T_{i}\ )\ \cdot\ c)$$

The MDEK1001 Kit is designed to test the real-time location system (RTLS) sensor modules of Decawave's DWM1001. Based on the DW1000 UWB transceiver IC, which incorporates UWB and Bluetooth antennas, RF circuitry, Nordic nRF52832 semiconductor and motion sensor, the DWM1001 board.

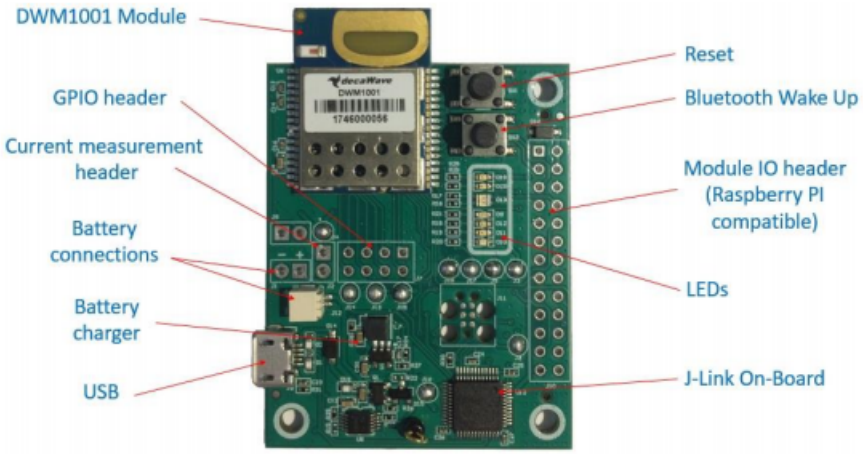


Figure 3.7 – The main components of the module development board DWM1001 [20]

The MDEK1001 kit includes: 12 units based on DWM1001 module/development board, embedded location stack (binary), gateway firmware and PC/Android app (in source code). These sensor modules can be configured as 'Anchors' or as 'Tags' using the Android app. While an 'Anchor' represents one of the fixed nodes in the system, a 'Tag' represents one a mobile node in the system. MDEK1001 operates on UWB channel 5 (6.5 GHz center frequency), delivering a data rate of up to 6.8 Mbps and a 30 m range. The kit is capable of updating the position of each tag on the network at a frequency of up to

10 Hz as a real-time solution.

The Decawave ranging system, MDEK1001, estimates the tags locations using trilateration. This mathematical technique uses the estimated rage measurements between the target location (tag) and the closest three known locations (anchors) to determine the position of the tag relative to the known locations of the anchors. For the range measurements used by the MDEK1001 RTLS during trilateration, the UWB tag uses the two-way-Time Of Flight (TOF) algorithm to link and estimate the distance to the nearest four anchors.

3.3.2 Setup and measurement results

MDEK1001's simple positioning configuration includes four anchors (preferably a rectangular one) to be positioned in the corners of the covered area, the tag which is to be located, and the Android app installed on a smartphone or tablet used to configure the main parameters of the nodes in the RTLS network. The anchors were placed in the corners of a 4.5 m x 4 m rectangle (measurement area), at a height of 1.8 m, on walls. Because of the polarisation of the antenna, the anchors were vertically oriented, facing one another. We were trying different positions of the tag within the room and outside to try to get results in LoS conditions and in NLOS. Anchor 1 is the Initiator and represents the origin of the system (0,0). Going counter clockwise, Anchor 2 is at (4.5 m, 0), Anchor 3 is at (4.5 m, 3 m), and Anchor 4 is at (0, 3 m). An Auto Positioning feature is provided by the Android RTLS application for MDEK1001 that enables the anchors to evaluate their positions by themselves.
This function was tested, resulting in: Anchor 1 (0,0), Anchor 2 (4.47 m, 0), Anchor 3 (4.48 m, 3.05 m), and Anchor 4 (0, 2.98 m). However, the proper positions of the anchors were reset manually.

Ranging starts automatically once devices have been added to the network. It's possible to see the anchors and the tag localized on a grid through choosing the grid option. The system allows to estimate the distance between each anchor and the target. The experimentation consisted in moving one mobile node to different ground-truth positions. This experimentation was repeated three times for each position. A map of the testing

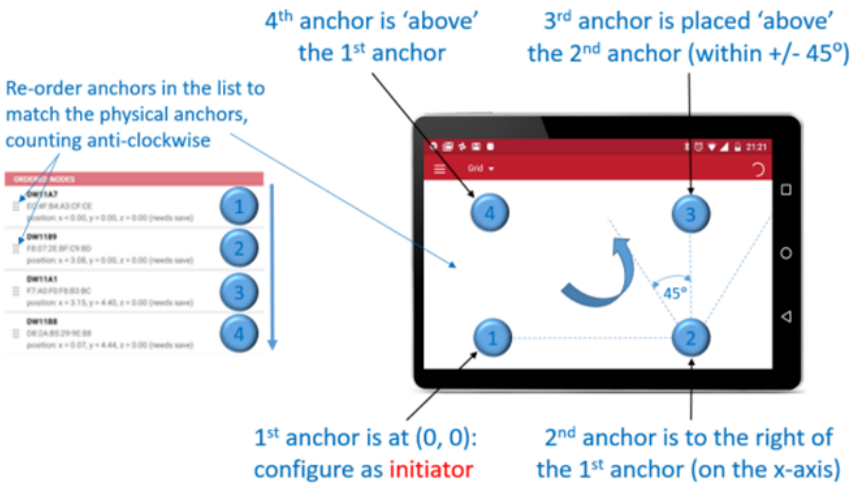


Figure 3.8 – Anchors positions

area is shown in the next figure, where it can be seen the floor plan with the detailed anchor positions and the distribution of the different test points. A transition from a position to the next one was done manually and random so to guarantee a good ground truth.

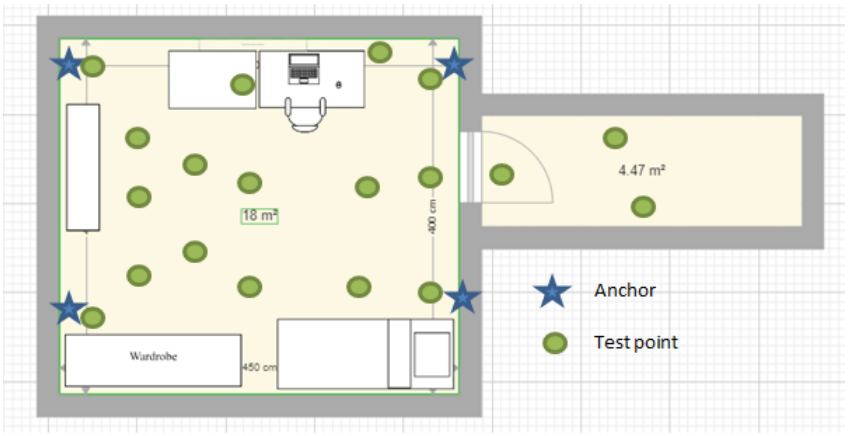


Figure 3.9 – Map of the room with the test points

3.3.3 Ranging Error

Propagation effects, clock drifts in nodes, and interference with other external radio sources are the potential sources of error in the UWB range. Owing to the propagation

of signals within blocking materials, the propagation effects can be triggered by extreme multipaths and propagation delays. UWB is capable of correctly calculating the LoS time of arrival in the case of moderate multipaths with a poor yet detectable direct path. However the measured ranges refer to the shorter NLoS path when the direct path is blocked, which may involve a significant delay from a few decimeters to several meters. In order to compare the theoretical expected error values with the experimental ones, we plotted the relationship among the real and measured distances of the system in the figure below.

The estimated points seem to be almost identical to the theoretical ones. However some outliers appear in the range of [0, 0.25 m]. In fact Decawave has shown some difficulties when detecting targets which are close to anchor nodes, precisely when target node is closer to the anchor more than 25 cm. Some outliers also are due to NLoS and the presence of obstacles. When absence of direct communication and when the distance between the tag and the anchor surpasses three meters the ranging error reaches sometimes 30 cm.

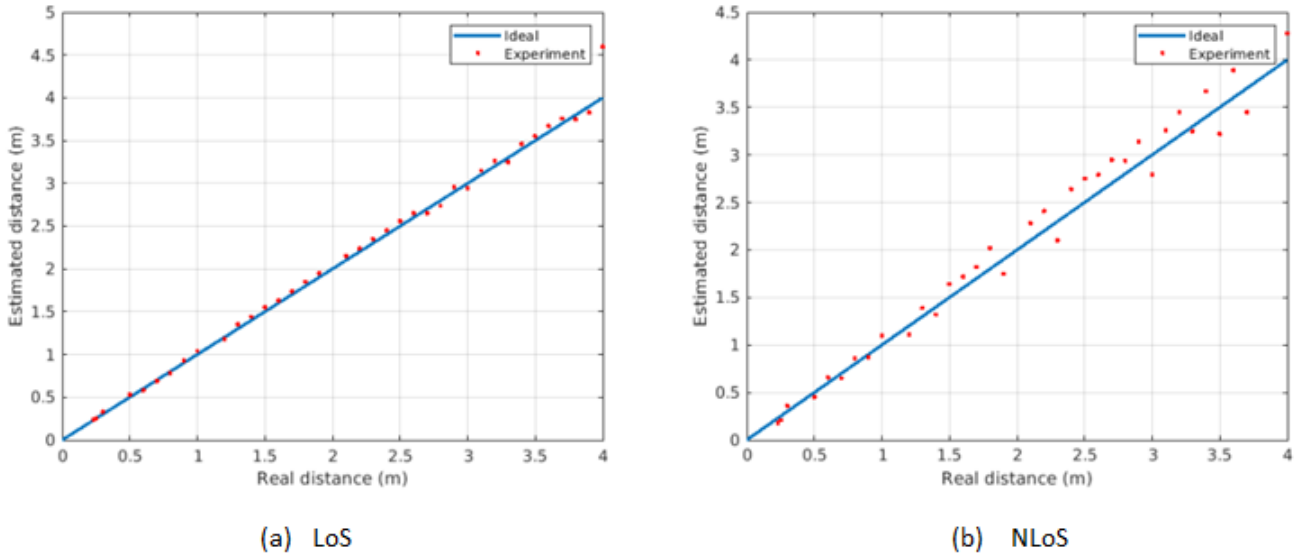


Figure 3.10 – Real versus estimated distances using Decawave system

3.4| Establishing the connection between the MSP430 launchpad and the Decawave module

In this section, we are presenting the communication protocols between the MSP430 and the DWM1001. Apart from using the DWM1001 module itself, the DWM1001 module can be physically connected to external controlling hardware, either wired or wirelessly over Bluetooth radio. The Positioning and Networking stack (PANS) API provides the users with four sets of API to call the PANS library functions through different interfaces. We have chosen UART for being the easiest solution in terms of wiring and because we don't need a synchronisation in our communication.

3.4.1 What is UART?

UART stands for Universal Asynchronous Receiver/Transmitter. It's not a communication protocol like SPI and I2C, but a physical circuit in a microcontroller, or a stand-alone IC. A UART's main purpose is to transmit and receive serial data. One of the best things about UART is that it only uses two wires between devices to transfer data. The concepts behind UART are simple to comprehend.

In UART communication, two UARTs interact directly with each other. The transmitting UART translates parallel data from a controlling system such as a CPU into serial form, transmits it to the receiving UART in serial form, and then converts the serial data back into the receiving device's parallel data. For transmitting data between two UARTs, only two wires are required. Data flows from the transmitting UART's Tx pin to the receiving UART's Rx pin:

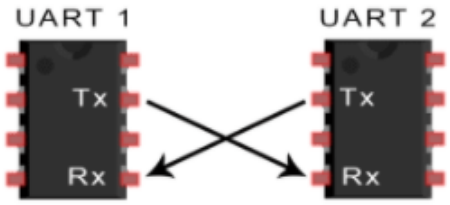


Figure 3.11 – UART communication protocol's wiring

UARTs transmit data asynchronously, which means that the output of bits from the transmitting UART to the sampling of bits by the receiving UART is synchronized with no clock signal. The transmitting UART adds start and stop bits to the data packet being transmitted instead of a clock signal. The beginning and end of the data packet are specified by these bits, so the UART receiver knows when to start reading the bits.

When a start bit is detected by the receiving UART, the incoming bits begin to be read at a particular frequency known as the baud rate. The Baud rate is a measure of data transmission speed, expressed in bits per second (bps). Both UARTs must work at approximately the same baud rate. Before the timing of bits becomes too far off, the baud rate between the transmitting and receiving UARTs can only vary by about 10 percent.

3.4.2 Access to the Decawave module using UART

The DWM1001 DEV board provides UART access through both the USB connector and the pins on the external connector. External header pins provided on J10 of the DWM1001 DEV board also provides the UART interface. The figure below shows the pins needed in the UART connection

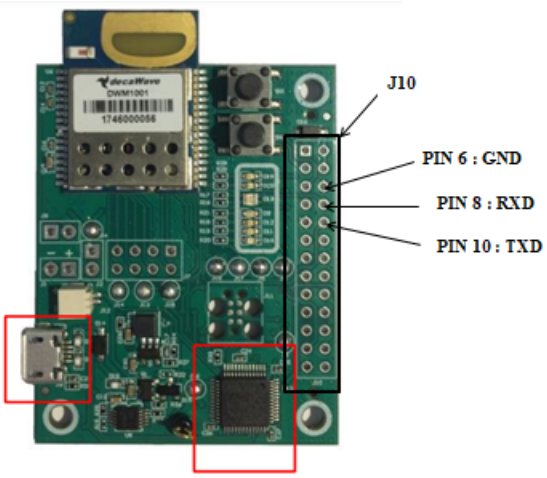


Figure 3.12 – DWM1001 pins dedicated to the UART communication [20]

The UART interface can be accessed by means of two modes, Shell and Generic. It's possible to use both Generic mode and Shell mode to communicate via the UART link

with the DWM1001 module. Generic mode is the default mode of the DWM1001 UART. The two modes are transferable:

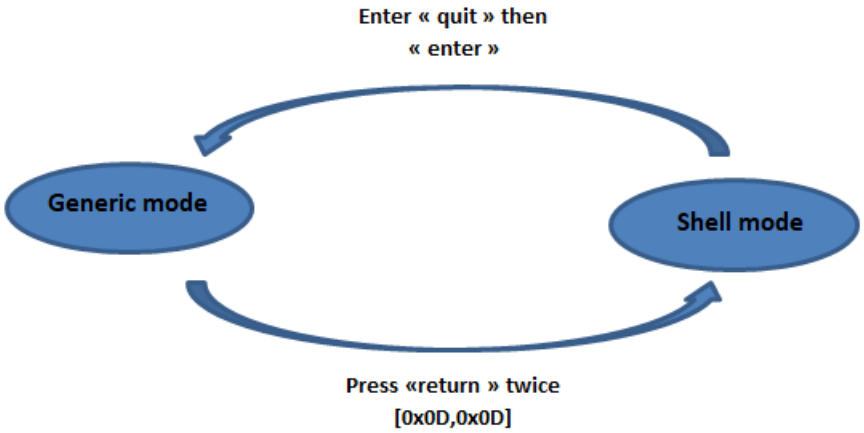


Figure 3.13 – Switching between Generic and Shell mode

3.4.3 UART communication using TLV

Host device and the DWM1001 module communicate using TLV requests/responses. TLV format, the Type-Length-Value encoding, is used in the DWM1001 module API. Data in TLV format always begins with the type byte, followed by the length byte, and then followed by a variable number of value bytes [0 to 253] as specified by the length byte.

3.4.4 Serial Communication within MSP430

In MSP430, serial communication is handled by an on chip peripheral called USCI (Universal Serial Communications Interface). The peripheral is designed in such a way that it can handle multiple serial communication formats , synchronous as well as asynchronous like SPI, I2C, IrDA, UART etc.MSP430G2553 has two USCI modules names as USCI_A0 and USCI_B0 for handling multiple communication formats. USCI_A0 can be configured to handle LIN, IrDA, SPI and asynchronous serial communication (UART) while USCI_B0 can handle SPI and I2C.

3.4.4.1 UART's Data Frame structure

If the transmitter requires data to be transmitted to another computer. By making a transition from high to low signal, it sends the start signal and the receiver recognizes this start bit and starts to receive data bit by bit. For example, the data in the image below is 8 bits. UART means we can submit just one character per frame with 8 bits.

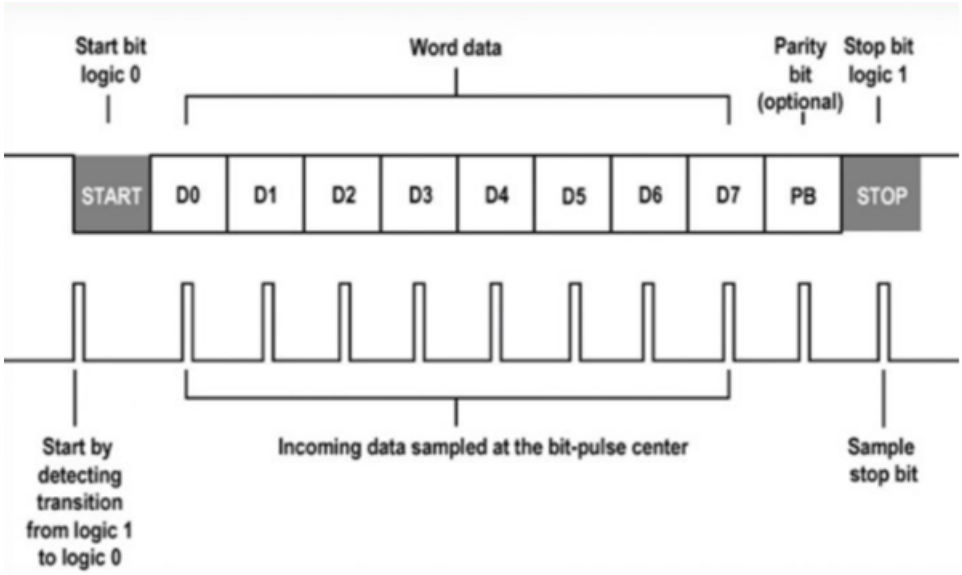


Figure 3.14 – UART Data Frame[22]

- **Start bit:** A frame always begins with a start bit (1 bit), this is used so that the communication partner recognizes after a pause when the communication starts.
- **Data:** The actual data record is always 7 or 8 bits long at TI. The LSB (Least Significant Bit) is sent first.
- **Parity:** The parity bit is used as a control to catch possible transmission errors. It's possible to choose between none (no parity bit used), even (the sum of the ones in the data record results in an even number) or odd (the sum of the ones in the data record results in an odd number).
- **Stop bit:** The frame is ended with 1 or 2 stop bits, which are used exclusively to synchronize the messages. In the example, 1 stop bit is used.

- **Baud rate:** Finally, the baud rate, i.e. the speed at which the frames are transmitted, must be set. TI offers typical baud rate values for different frequencies on the next figure.

3.4.4.2 Configuring the Baud rate

MSP430 data sheet has an in depth description about configuring the baud rate for various modes. Easiest way to configure the baud rate of a MSP430 device is to use the Baud rate selection table listed in the user manual.

USCI Operation: UART Mode

Table 15–4. Commonly Used Baud Rates, Settings, and Errors, UCOS16 = 0

BRCLK frequency [Hz]	Baud Rate [Baud]	UCBRx	UCBRSx	UCBRFx	Max. TX Error [%]		Max. RX Error [%]	
32,768	1200	27	2	0	–2.8	1.4	–5.9	2.0
32,768	2400	13	6	0	–4.8	6.0	–9.7	8.3
32,768	4800	6	7	0	–12.1	5.7	–13.4	19.0
32,768	9600	3	3	0	–21.1	15.2	–44.3	21.3
1,048,576	9600	109	2	0	–0.2	0.7	–1.0	0.8
1,048,576	19200	54	5	0	–1.1	1.0	–1.5	2.5
1,048,576	38400	27	2	0	–2.8	1.4	–5.9	2.0
1,048,576	56000	18	6	0	–3.9	1.1	–4.6	5.7
1,048,576	115200	9	1	0	–1.1	10.7	–11.5	11.3
1,048,576	128000	8	1	0	–8.9	7.5	–13.8	14.8
1,048,576	256000	4	1	0	–2.3	25.4	–13.4	38.8
1,000,000	9600	104	1	0	–0.5	0.6	–0.9	1.2
1,000,000	19200	52	0	0	–1.8	0	–2.6	0.9
1,000,000	38400	26	0	0	–1.8	0	–3.6	1.8
1,000,000	56000	17	7	0	–4.8	0.8	–8.0	3.2
1,000,000	115200	8	6	0	–7.8	6.4	–9.7	16.1
1,000,000	128000	7	7	0	–10.4	6.4	–18.0	11.6
1,000,000	256000	3	7	0	–29.6	0	–43.6	5.2

Figure 3.15 – Baud rate configuration[21]

For example to provide a baud rate equal to 9600 bps the registers UCBRx and UCBRSx should be set to 104 and 1 respectively.
The next formula allows to choose correctly the value of the register UCA0BRx in case we don't want the predefined values in the table above.

$$UCA0BRx = INT(F_{CPU} / BAUD_{SOLL})$$

- **F_{CPU}:** stands for the frequency with which the UART module runs.
- **$BAUD_{SOLL}$:** is the desired baud rate at which UART can send and receive information
- **INT:** This operand should ensure that only integers can be used.

3.4.4.3 Setting the launchpad up for the UART connection

Now let's take a quick look at the hardware (Launchpad). As already mentioned, pins 1.1 and 1.2 are responsible for UART. The Launchpad has a USB to UART converter and can therefore be connected directly to the PC. The launchpad's supply cable can also be used for UART communication. To do this, the jumpers for the hardware UART must be plugged in (see the figure below - HW-UART).

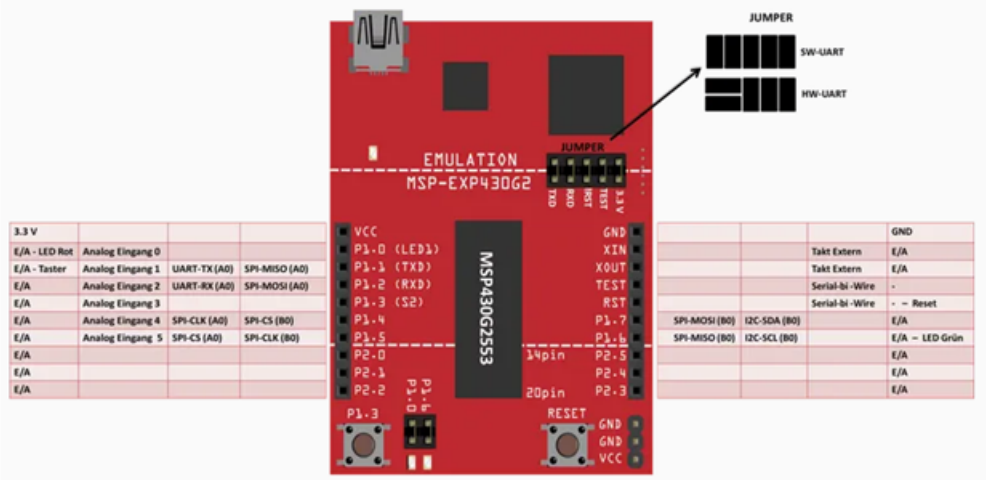


Figure 3.16 – Setting the launchpad jumpers in UART mode[22]

Chapter 4

Realization and results

4.1| Introduction

The approach of energy conservation fixed, and after conceiving the system we will present in this chapter the tasks elaborated during the internship including the development environment, the implementation and the achieved results.

4.2| Tools and development environment

In this section we are presenting the development environment used to settle the system up. The softwares used to interact with the different components and to ensure the communication between them are mentioned below.

4.2.1 Code composer studio

Code Composer Studio is an integrated development (IDE) environment that supports the portfolio of TI Microcontroller and Embedded Processors. A suite of tools used for developing and debugging embedded applications is included in Code Composer Studio. This includes a C/C++ compiler, source code editor, project build environment, debugger, profiler, and several other features that are optimized.

The intuitive IDE offers a single user interface that takes user through every step of the flow of application development. Familiar tools and interfaces allow users to get started more easily than ever. Code Composer Studio blends the benefits of the Eclipse software framework with TI's advanced embedded debug capabilities, resulting in a compelling environment for embedded developers that is rich in features.

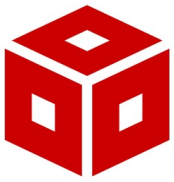


Figure 4.1 – Code composer studio

We will use Code composer studio to program the MSP430G2553 launchpad and to send UART messages to the DWM1001.

4.2.2 SEGGER Embedded Studio for ARM

SEGGER Embedded Studio for ARM is a streamlined integrated development environment, compilation tools, and libraries for building, testing, and deploying applications on ARM and Cortex microcontrollers.



Figure 4.2 – SEGGER Embedded Studio

This development environment will play the role of the interface with the DWM1001. Configuration and access to the module are ensured by this latter.

4.2.3 Hterm

Even though USB can now be found on almost all devices and often replaces the old RS232 interfaces, for example for keyboard and computer mouse, the RS232 interface is

still used. One tool that can deal with this is HTerm.



Figure 4.3 – Hterm

HTerm by Tobias Hammer is a terminal program for the RS232 serial interface and is aimed to programmers and makers who use it. The software for Windows and Linux can be used to address a device for controlling technical processes, a microcontroller such as the Arduino and, in particular, the UART interface. HTerm is not a terminal program in the usual sense for the terminal. In the graphical user interface it's possible to specify, among other things, the port such as COM 1, the baud rate and the parity for sending and receiving. In addition, there is an input and output for decimal, hexadecimal and binary numbers as well as ASCII in the interface, whereby the hexadecimal, binary, decimal and ASCII representation makes HTerm a tool for debugging and editing. Various newline codes and font sizes can also be selected.

4.3| Results

To confirm our work and prove the utility of the energy efficiency approach based on wake-up receiver we present some measures obtained during the test phase highlighting the current consumption within the devices, the UART communication between them to end up finally with a general conclusion resuming the different parts of the work and to analyze the work achieved.

4.3.1 UART communication

In this stage of the project, we aimed to work on the MSP430 launchpad and to manipulate its registers in order to output a UART message that allows to set DWM1001

into sleep mode when absence of localization requests. For this reason we took advantage from the interrupts to use the switch button to trigger the sleep request. Before starting the implementation, we should set the scheme up by establishing the right wiring between both printed circuit boards (PCBs)

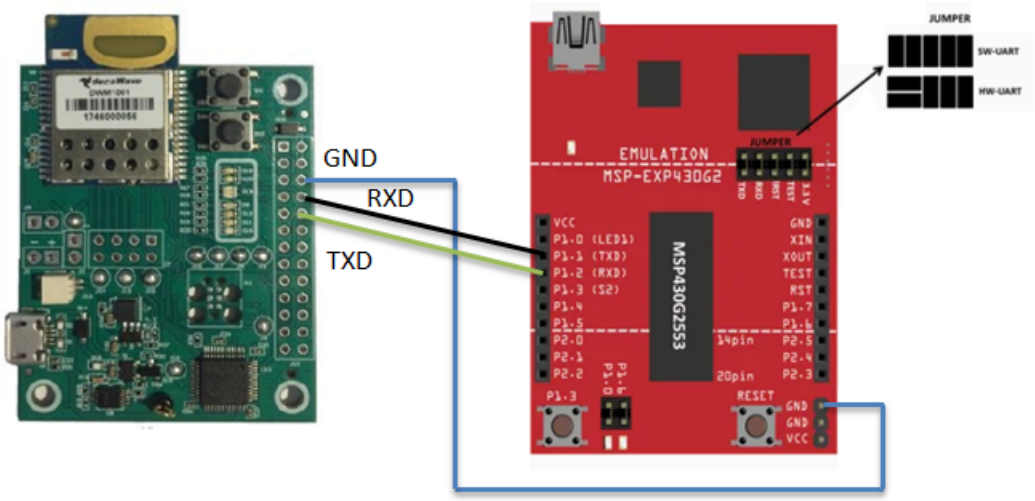


Figure 4.4 – UART wiring

As shown in the previous figure, the wires should be set in a way that the receiver of the DWM1001 is connected to the transceiver of the launchpad, its transceiver is connected to the receiver of the MSP430 and both GNDs are connected together. The jumpers of the TXD and RXD pins should be mounted in a parallel position to ensure the progress of the UART communcation. The two PCBs are set up, so the next phase was the implementation of the MSP430 code that sends UART messages to the DWM1001 that consists in reset/sleep requests. To visualize the interaction we used the Hterm. The results are presente below :
According to the Firmware user guide of the DWM1001, these requests should be sent in a TLV format as mentionned before. The messages are mentionned with the expected response received from the DWM1001. For the example of the sleep request here are shown the needed commands.

TLV request	
Type	Length
0x0A	0x00

TLV response		
Type	Length	Value
		err_code
0x40	0x01	0x00

Figure 4.5 – Sleep commands

After executing the mentioned command here is the result obtained.

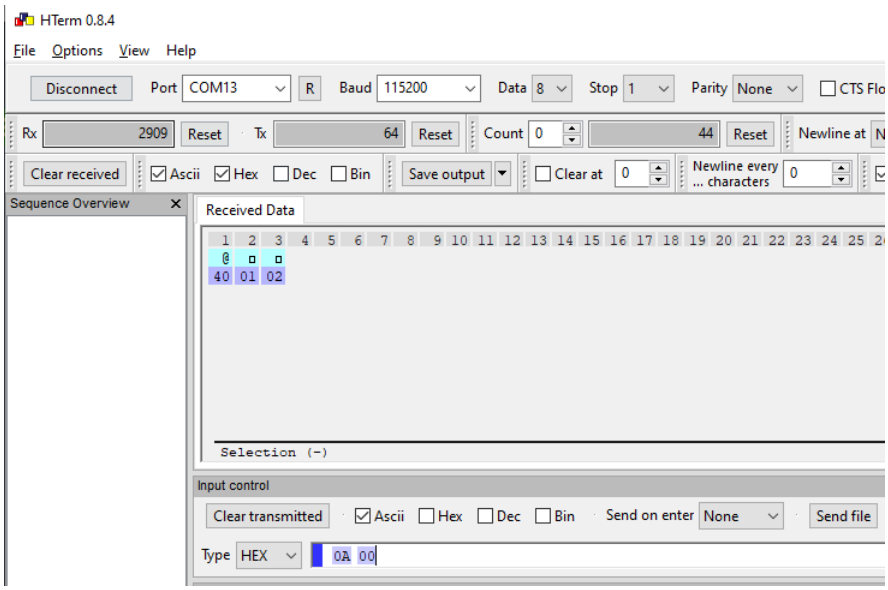


Figure 4.6 – Sleep request and response

As shown in the figure, the response obtained is compliant to the given result in the Firmware guide so we have established a successful UART communication between the launchpad and the DWM1001. In the next section we are presenting the current consumption results in each of the components since its our major concern in this work. So we are aiming to reduce current consumption within the hole circuit in order to ensure a longevity of the battery .

4.3.2 Energy consumption

Wake-up receivers (WuRx) are intended to replace the main transceivers for idle listening tasks while consuming much less power. For extremely low latency communication, WuRx has to remain always on. If an ultra-low energy consumption can be realized, the

sender network saves more energy and can last longer than a network using a duty-cycle scheme. Therefore it's important to guarantee a minimum of energy consumption within every component of our proposed model. As a start point, the launchpad was our main concern so we took advantage of the low power modes included in this latter. At the first place the current consumption was relatively high around 85 µA. MSP430 is designed for ultra-low power applications. It offers the lowest power consumption and the perfect mix of integrated peripherals for a variety of battery operated applications. MSP430G2553 microcontroller has low power modes(LPM) 0, 1, 2, 3 and 4. Power consumption in various modes is shown in the next figure:

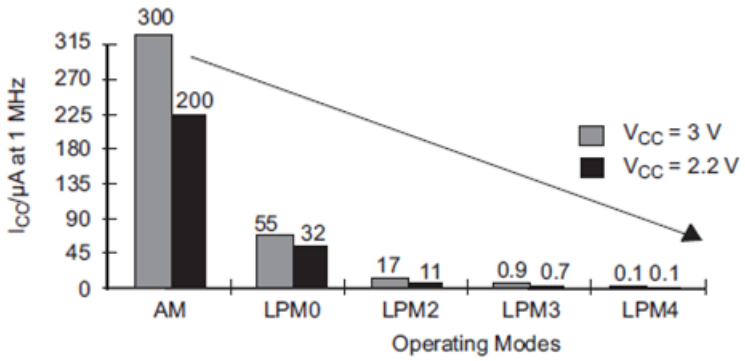


Figure 4.7 – Low power modes in MSP430 [17]

As LPM 4 corresponds to the lowest current that's why we managed to proceed with it. In this mode all the clocks including ACLK are disabled in this mode. The controller can only be wakened from this mode by an external interrupt. This mode consumes the least power compared to all other modes. After configuring the MSP430 to LPM4 we wanted to make sure that the current consumption in the PCB is identical to what's mentioned in its theoretical presentation. The figure below shows that we got the exact measurement and the energy consumption is set to its minimum.

After fixing the current consumption in the MSP430, the next step was ensuring that the energy consumption remains at low levels after linking both PCBs (MSP430 launchpad and DWM1001). As we mentioned, our work was based on keeping the Decawave module into sleep mode to reach our goal. Therefore in this point of the experiment we are measuring the current state after the UART connection is established between the two components.

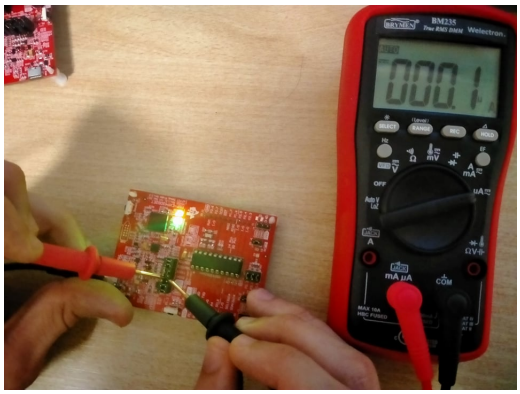


Figure 4.8 – Current measurement in MSP430's PCB using LPM4

The obtained result was 65 µA. This result seems to be a bit high concerning to what we obtained when the current was measured on the MSP430 independently.

4.3.3 Implementation of the WuRx

The communication between the MSP430 launchpad and the DWM1001 is established, the next step was integrating the wake-up receiver within the circuit. The wake up procedure is divided in two parts : receiving and transmission. Therefore both transceiver and receiver of the wake up module are programmed separately. The transmitter's program consists in sending a wake-up package on button press (P1.3 of the MSP430) through UART connection. The program uses sleep mode on SPIRIT1 and LPM4 on MSP430 so the total standby current is about 10 µA. It also connects antenna to right SMA connector (SPIRIT1 RF connection). The receiver's program uses the SPI connection since the UART is busy with the transceiver's program. It uses the module without SPIRIT1 and connects antenna to left SMA connector (WuRX RF connection). The scenario goes as mentioned above : the wake-up receiver gets the localization request, a wake-up package is sent then to the MSP430, finally a message is sent by this latter to the Decawave module to wake it up, localize the target node and then goes back to sleep after a certain period.

Current consumption within the circuit has been measured during the experiment to discuss later the reached results.

State	Current measurement
Deep sleep	60 µA
Wake up	42.3 mA
Idle	19.8 mA
Transmission	153.4mA
Reception	175 mA

Table 4.1 – Current measurement in different phases of the experiment

4.4| Results interpretation

Results show that in sleep mode, which corresponds to deep sleep in DWM1001, the current inside the circuit reaches its minimum (60 µA). When the wake-up packet is sent from the Wake-up receiver to the MSP430 the current increases significantly to reach 42.3 mA and then stabilizes around 20 mA in idle mode. This difference in current levels is due mainly to the active components of the Decawave module. In fact in sleep mode all the peripherals are set to their lower power consumption whereas in idle mode the MCU is awake. In transmission and reception phases, when the messages of wake-up are communicated to the DWM1001 and the localization measures are transmitted, the current measurement goes again high to reach 175 mA. Based on these results we can deduct that the main cause of energy consumption within the sensor node basically are its components when they're set into idle mode or data transmission which is already mentioned in the previous section. Setting the DWM1001 into sleep mode when there are no localization requests seems to be a good choice and energy efficient.

Conclusion and Future Work

Through this work we considered the issue of energy conservation and battery lifetime extension of wireless sensor nodes in indoor localization platforms. In particular, we propose an on-demand indoor localization platform using sensor nodes with built-in Wakeup Receiver WuRx. The sensor nodes in the platform are kept in sleep mode the majority of the time consuming around 60 µA. This approach has shown high energy efficiency by saving up to 90% of the consumed energy compared to the duty-cycled approaches and the battery of the sensor nodes lifetime increases outstandingly especially in low-demand scenarios.The WuRx-based indoor localization approach also allows the platform to be alert and responsive to localization requests with the necessary minimum nodes and lower energy costs unlike the duty-cycled approaches that involve a denser network since certain nodes are in sleep mode and the request can not be detected. By setting four anchors on the ceiling of 8×8 m room, we have also tested the accuracy of the mentioned localization system and set the target in different positions within it. The variation in the position estimation is whether due to the distance between the target node and the anchors or the non-line of sight measurements. In fact obstacles such as walls, doors can affect the ultra wide band signals. The WuRx-based ODIL platform has provided an average accuracy of 15 cm using an anchor deployment density of 0.25 Anchor/m^2 .

Perspectives

The obtained results are very promising and show that the on demand protocol is beneficial for low power indoor localization. The first challenge ahead is to work on the conservation of energy. The results achieved are energy preserving but still consuming power. One of the main axis that we can consider is the communication between the

micro-controller and the decawave module which appeared to be greedy in terms of energy. In fact shutting down the communication while it's not needed might be an option. From another hand, the calibration process, which is often used in indoor range-based localization systems, has a significant impact on accuracy because the parameters extracted have an outstanding impact on the accuracy of the distance estimation between anchors and mobile targets .In addition, the values of these parameters are not constant and differ throughout the building. However, it won't be a very realistic and cost-effective job to manually estimate these parameters. For this purpose, in order to reduce the expense of the calibration step and to improve accuracy, we intend to develop a self-calibrating method for WSN-based indoor localization.

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Appendix A : Decawave module Datasheet

A.1 Key features

Ranging accuracy	within 10cm
UWB Channel	6.5 GHz
Data rate	6.8 Mbps
Line-of-sight range	60 m
UWB compliant	IEEE 802.15.4-2011
Connectivity	Bluetooth
Chip antenna	Bluetooth
Motion sensor	3-axis accelerometer
Supply voltage	2.8 V to 3.6 V
Size	19.1 mm x 26.2 mm x 2.6 mm

Table A.1: Decawave module DWM1001 technical specifications

A.2 Key benefits

- Enables anchors, tags gateways to quickly get an entire RTLS system up-and-running.
- Accelerates product designs for faster Time-to-Market reduced development costs.
- Ready-to-go embedded firmware to minimise software development.
- Over-the-air updates.
- User API to DWM1001 firmware (available as a library) for user code customisation.

- On-board Bluetooth SMART for connectivity to phones/tablets/PCs.
- SPI, UART and Bluetooth APIs to access DWM1001 firmware from an external device.
- Low-power hardware design and software architecture for longer battery life.

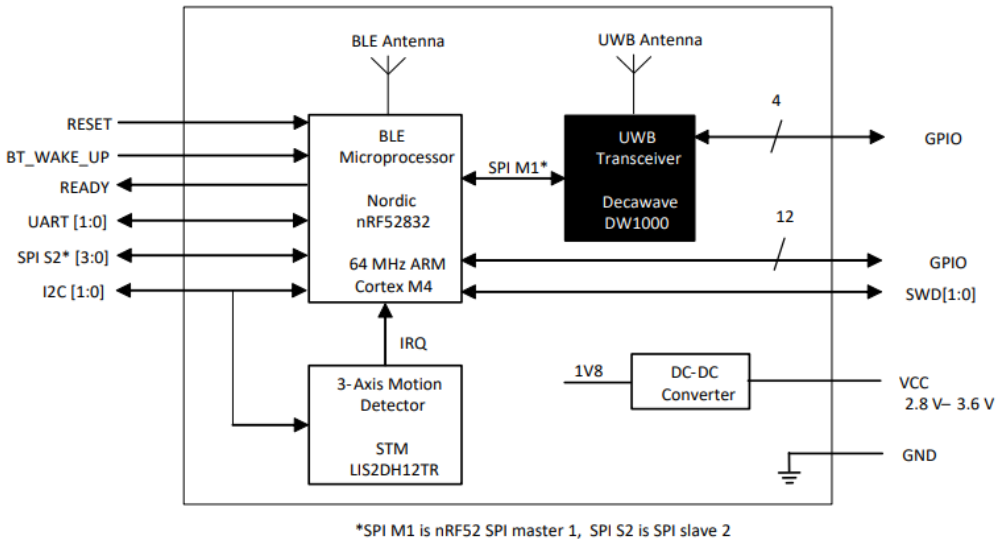


Figure A.1: High level block diagram

Appendix B : MSP430G2553 Datasheet

MIXED SIGNAL MICROCONTROLLER

FEATURES

- Low Supply-Voltage Range: 1.8 V to 3.6 V
- Ultra-Low Power Consumption
  - Active Mode: 230 µA at 1 MHz, 2.2 V
  - Standby Mode: 0.5 µA
  - Off Mode (RAM Retention): 0.1 µA
- Five Power-Saving Modes
- Ultra-Fast Wake-Up From Standby Mode in Less Than 1 µs
- 16-Bit RISC Architecture, 62.5-ns Instruction Cycle Time
- Basic Clock Module Configurations
  - Internal Frequencies up to 16 MHz With Four Calibrated Frequency
  - Internal Very-Low-Power Low-Frequency (LF) Oscillator
  - 32-kHz Crystal
  - External Digital Clock Source
- Two 16-Bit Timer_A With Three Capture/Compare Registers
- Up to 24 Capacitive-Touch Enabled I/O Pins
- Universal Serial Communication Interface (USCI)
  - Enhanced UART Supporting Auto Baudrate Detection (LIN)
  - IrDA Encoder and Decoder
  - Synchronous SPI
  - I^2C™
- On-Chip Comparator for Analog Signal Compare Function or Slope Analog-to-Digital (A/D) Conversion
- 10-Bit 200-ksps Analog-to-Digital (A/D) Converter With Internal Reference, Sample-and-Hold, and Autoscan (See Table 1)
- Brownout Detector
- Serial Onboard Programming, No External Programming Voltage Needed, Programmable Code Protection by Security Fuse
- On-Chip Emulation Logic With Spy-Bi-Wire Interface
- Family Members are Summarized in Table 1
- Package Options
  - TSSOP: 20 Pin, 28 Pin
  - PDIP: 20 Pin
  - QFN: 32 Pin

Figure B.1: MSP430's Features

Device	BSL	EEM	Flash (KB)	RAM (B)	Timer_A	COMP_A+ Channel	ADC10 Channel	USCI_A0, USCI_B0	Clock	I/O	Package Type
MSP430G2553IRHB32	1	1	16	512	2x TA3	8	8	1	LF, DCO, VLO	24	32-QFN
MSP430G2553IPW28										24	28-TSSOP
MSP430G2553IPW20										16	20-TSSOP
MSP430G2553IN20										16	20-PDIP
MSP430G2453IRHB32	1	1	8	512	2x TA3	8	8	1	LF, DCO, VLO	24	32-QFN
MSP430G2453IPW28										24	28-TSSOP
MSP430G2453IPW20										16	20-TSSOP
MSP430G2453IN20										16	20-PDIP
MSP430G2353IRHB32	1	1	4	256	2x TA3	8	8	1	LF, DCO, VLO	24	32-QFN
MSP430G2353IPW28										24	28-TSSOP
MSP430G2353IPW20										16	20-TSSOP
MSP430G2353IN20										16	20-PDIP
MSP430G2253IRHB32	1	1	2	256	2x TA3	8	8	1	LF, DCO, VLO	24	32-QFN
MSP430G2253IPW28										24	28-TSSOP
MSP430G2253IPW20										16	20-TSSOP
MSP430G2253IN20										16	20-PDIP
MSP430G2153IRHB32	1	1	1	256	2x TA3	8	8	1	LF, DCO, VLO	24	32-QFN
MSP430G2153IPW28										24	28-TSSOP
MSP430G2153IPW20										16	20-TSSOP
MSP430G2153IN20										16	20-PDIP
MSP430G2513IRHB32	1	1	16	512	2x TA3	8	-	1	LF, DCO, VLO	24	32-QFN
MSP430G2513IPW28										24	28-TSSOP
MSP430G2513IPW20										16	20-TSSOP
MSP430G2513IN20										16	20-PDIP
MSP430G2413IRHB32	1	1	8	512	2x TA3	8	-	1	LF, DCO, VLO	24	32-QFN
MSP430G2413IPW28										24	28-TSSOP
MSP430G2413IPW20										16	20-TSSOP
MSP430G2413IN20										16	20-PDIP
MSP430G2313IRHB32	1	1	4	256	2x TA3	8	-	1	LF, DCO, VLO	24	32-QFN
MSP430G2313IPW28										24	28-TSSOP
MSP430G2313IPW20										16	20-TSSOP
MSP430G2313IN20										16	20-PDIP
MSP430G2213IRHB32	1	1	2	256	2x TA3	8	-	1	LF, DCO, VLO	24	32-QFN
MSP430G2213IPW28										24	28-TSSOP
MSP430G2213IPW20										16	20-TSSOP
MSP430G2213IN20										16	20-PDIP

Figure B.2: Table 1

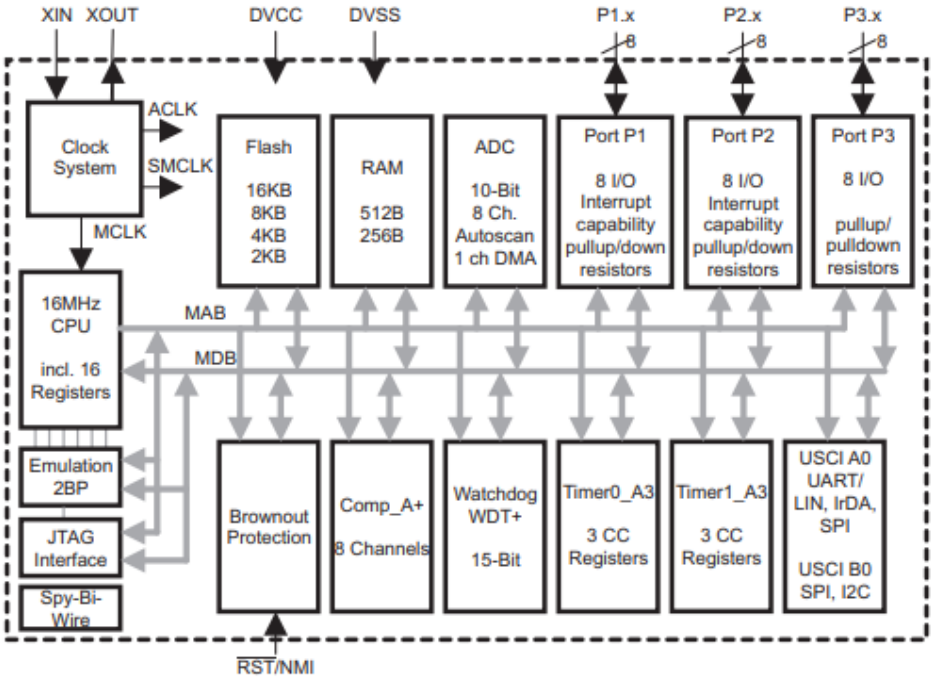


Figure B.3: Functional Block Diagram, MSP430G2x53

Appendix C : Decawave's localization measurements

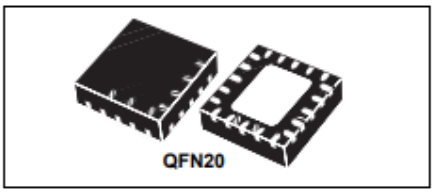
Real distance	Estimated	Error
0.05	NA	NA
0.075	NA	NA
0.1	NA	NA
0.15	NA	NA
0.2	NA	NA
0.23	0.24	0.01
0.25	0.25	0.0
0.30	0.33	0.03
0.5	0.53	0.03
0.6	0.58	0.02
0.7	0.69	0.01
0.8	0.78	0.02
0.9	0.93	0.03
1.0	1.04	0.04
1.2	1.18	0.02
1.3	1.35	0.05
1.4	1.44	0.04
1.5	1.55	0.05
1.6	1.63	0.03
1.7	1.74	0.04
1.8	1.85	0.05
1.9	1.95	0.05
2.1	2.15	0.05
2.2	2.24	0.04
2.3	2.35	0.05
2.4	2.45	0.05
2.5	2.56	0.06
2.6	2.65	0.05
2.7	2.65	0.06
2.8	2.74	0.06
2.9	2.96	0.06
3	2.94	0.06
3.1	3.15	0.05
3.20	3.26	0.06
3.3	3.25	0.05
3.4	3.46	0.06
3.5	3.55	0.05
3.6	3.67	0.07
3.7	3.76	0.06
3.8	3.75	0.05
3.9	3.83	0.07
4	4.6	0.06

Figure C.1: Ranging measurements in LoS

Real distance	Estimated	Error
0.05	NA	NA
0.075	NA	NA
0.1	NA	NA
0.15	NA	NA
0.2	NA	NA
0.23	0.18	0.05
0.25	0.21	0.04
0.30	0.36	0.06
0.5	0.45	0.05
0.6	0.66	0.06
0.7	0.65	0.05
0.8	0.86	0.06
0.9	0.87	0.03
1.0	1.10	0.1
1.2	1.11	0.09
1.3	1.39	0.09
1.4	1.32	0.08
1.5	1.64	0.09
1.6	1.72	0.12
1.7	1.82	0.12
1.8	2.02	0.22
1.9	1.75	0.15
2.1	2.28	0.18
2.2	2.41	0.21
2.3	2.10	0.20
2.4	2.64	0.24
2.5	2.75	0.25
2.6	2.79	0.19
2.7	2.95	0.25
2.8	2.94	0.14
2.9	3.14	0.24
3	2.79	0.21
3.1	3.26	0.26
3.20	3.45	0.25
3.3	3.25	0.05
3.4	3.67	0.26
3.5	3.22	0.28
3.6	3.89	0.29
3.7	3.45	0.25
3.8	4.05	0.25
3.9	4.29	0.3
4	4.28	0.28

Figure C.2: Ranging measurements in NLoS

Appendix D : SPIRIT1 Datasheet



Features

- Frequency bands: 150-174 MHz, 300-348 MHz, 387-470 MHz, 779-956 MHz
- Modulation schemes: 2-FSK, GFSK, MSK, GMSK, OOK, and ASK
- Air data rate from 1 to 500 kbps
- Very low power consumption (9 mA RX and 21 mA TX at +11 dBm)
- Programmable RX digital filter from 1 kHz to 800 kHz
- Programmable channel spacing (12.5 kHz min.)
- Excellent performance of receiver sensitivity (-118 dBm), selectivity, and blocking
- Programmable output power up to +16 dBm
- Fast startup and frequency synthesizer settling time (6 µs)
- Frequency offset compensation
- Integrated temperature sensor
- Battery indicator and low battery detector
- RX and TX FIFO buffer (96 bytes each)
- Configurability via SPI interface
- Automatic acknowledgment, retransmission, and timeout protocol engine
- AES 128-bit encryption co-processor
- Antenna diversity algorithm
- Fully integrated ultra low power RC oscillator
- Wake-up on internal timer and wake-up on external event
- Flexible packet length with dynamic payload length
- Sync word detection
- Address check
- Automatic CRC handling
- FEC with interleaving
- Digital RSSI output
- Programmable carrier sense (CS) indicator
- Automatic clear channel assessment (CCA) before transmitting (for listen-before-talk systems). Embedded CSMA/CA protocol
- Programmable preamble quality indicator (PQI)
- Whitening and de-whitening of data
- Wireless M-BUS, EN 300 220, FCC CFR47 15 (15.205, 15.209, 15.231, 15.247, 15.249), and ARIB STD T-67, T93, T-108 compliant
- QFN20 4x4 mm RoHS package
- Operating temperature range from -40 °C to 105 °C

Applications

- AMR (automatic meter reading)
- Home and building automation
- WSN (wireless sensors network)
- Industrial monitoring and control
- Wireless fire and security alarm systems
- Point-to-point wireless link

Table 1. Device summary

Order code	Package	Packing
SPIRIT1QTR	QFN20	Tape and reel

Figure D.1: SPIRIT1 : Features and applications